

Structure-Specific Representational Priors Causally Control the Grokking Delay

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Abstract

Grokking — generalization arriving long after training-set interpolation — has recently been shown to be controllable by structure-agnostic interventions: gradient filtering, weight-norm clamping, and geometric penalties on hidden representations. Whether the delay specifically measures the time to form *task-structured* representations has remained an observational claim. We test it causally. Training a one-layer transformer on modular addition, we add a supervised-contrastive auxiliary loss on hidden representations whose positive sets encode the task’s true equivalence structure ($c = (a + b) \bmod p$), and compare against an exactly-matched control whose positive sets encode a shuffled structure of identical class-size distribution, loss form, strength, and geometry. The true-structure prior generalizes in 22/30 runs — up to $2.75\times$ faster than baseline, the fastest transitions we observe, though the acceleration is dose-dependent (median per-seed speedup arises only at the strongest dose we test; weaker doses slow the median transition) — while the shuffled prior generalizes in 0/20 (Fisher exact $p = 1.3 \times 10^{-7}$). Because the auxiliary loss inflates the weight norm, and the norm alone is known to set the grokking timescale, we replay each intervention run’s exact norm trajectory onto plain cross-entropy training: 0/15 of these norm-matched controls generalize, collapsing instead into logit-scale saturation — so the acceleration is not norm-mediated. Representation probes (embedding Fourier concentration, class-cluster formation, CKA) show the true-structure prior builds class-clustered representations during the apparent plateau, ahead of and predictive of each run’s generalization time. The grokking delay is, causally, the time to form the right representational structure: injecting that structure can collapse the delay, and injecting the wrong structure at equal strength prevents generalization entirely.

1 Introduction

When a small transformer is trained on a modular arithmetic task, it memorizes the training set within a few hundred epochs, then sits at chance-level test accuracy for tens of thousands of epochs before abruptly generalizing — the phenomenon known as grokking [1]. The delay between training-set interpolation and generalization has become a model system for studying how neural networks transition from memorization to structured solutions, because in this setting the endpoint is fully understood: the generalizing network implements a specific Fourier multiplication circuit whose components can be measured as they form [2, 3].

A rapidly growing body of work shows that this delay is not a fixed property of the task but a controllable quantity. Filtering the gradient signal to amplify its slow component accelerates grokking by orders of magnitude [4, 5]; clamping the total weight norm rescales the time-to-grok along an exponential delay law [6], plausibly mediated by logit scale [7]; and penalizing the radial component of hidden activations — constraining representations to a hypersphere — yields up to

6.3 $\times$ acceleration [8]. These interventions differ in the knob they turn (optimizer, parameter norm, representation geometry), but they share a fundamental property: all are *structure-agnostic*. None of them tells the network anything about which structure the task actually requires.

Meanwhile, observational accounts increasingly locate the delay in representation formation. Chou et al. [9] decompose learning into a fast, quickly train-biased readout and a slow, continuously improving encoder, and argue that grokking occurs when representation quality catches up; Sivasankar [10] find that the synchronization of Fourier sub-circuits precedes the accuracy jump. If these accounts are right, the delay is, mechanistically, *the time it takes task-appropriate representational structure to form* — and injecting that structure directly should collapse the delay, while injecting equally strong but wrong structure should not. This causal prediction has not been tested: existing accelerators cannot test it, precisely because they are structure-agnostic.

We perform that test. We add a supervised-contrastive auxiliary loss [11] on the hidden representations of a one-layer transformer learning modular addition, and manipulate only the *content* of its positive sets: either the task’s true equivalence structure (examples sharing $(a + b) \bmod p$) or a shuffled control — a fixed random partition with identical class-size distribution, loss form, and strength. Because auxiliary losses also perturb the weight-norm trajectory, and the norm alone rescales the grokking timescale [6], we additionally run a weight-norm-matched control that replays the intervention’s norm trajectory onto standard training, plus a Grokfast baseline [4] as the reference optimization-side accelerator. Throughout training we track when representational structure forms — Fourier concentration of the embedding [2], class-clustering of held-out representations, and CKA trajectories — so that any change in the delay can be aligned against the timing of structure formation.

Our contributions:

- **Structure-specificity.** Across 10 seeds and three strengths, the true-structure prior generalizes in 22/30 runs, up to 2.75 $\times$ faster than the same seed’s baseline — the fastest transitions in our dataset — while the exactly-matched shuffled-structure prior generalizes in 0/20 (Fisher exact $p = 1.3 \times 10^{-7}$). The content of the injected structure, not the presence, strength, or geometry of the auxiliary loss, controls the delay.
- **Not norm-mediated.** The auxiliary loss inflates the total weight norm 2–3 $\times$; replaying each intervention run’s exact norm trajectory onto plain cross-entropy training yields 0/15 generalization, with collapse into logit-scale saturation. The structural prior generalizes at norms where structure-agnostic training cannot — bounding the scope of the weight-norm delay law [6] — and additionally reveals a structure-agnostic anti-saturation channel shared by both contrastive conditions.
- **Representation timing.** Class-cluster formation in held-out representations rises during the apparent plateau, ahead of each run’s accuracy jump, and its timing tracks generalization timing across conditions — the interventional counterpart of observational representation-first accounts of grokking [9, 10].
- **A bidirectional lever with a characterized failure mode.** The same intervention that produces the fastest generalization also stalls a minority of seeds (a race between injected structure and norm-driven saturation), and wrong structure at high strength delays even memorization — grokking dynamics can be steered in both directions by representational content alone.

2 Related Work

Grokking and its mechanisms. Grokking — generalization emerging long after the training set is interpolated — was first reported by Power et al. [1] in small transformers trained on modular arithmetic. Nanda et al. [2] reverse-engineered the generalizing solution for modular addition into a Fourier multiplication circuit and showed via progress measures that the circuit forms gradually during the apparent plateau, with the visible accuracy jump reflecting a later “clean-up” of the memorization component. Liu et al. [12] tied generalization to the formation of structured embeddings, Gromov [3] gave closed-form Fourier solutions for two-layer networks, Liu et al. [13] located grokking in the mismatch between initialization scale and dataset size, and Varma et al. [14] explained the transition through the greater parameter efficiency of the generalizing circuit under weight decay.

Structure-agnostic control of the delay. A rapidly growing 2026 literature shows the grokking delay is not fixed but controllable — so far, exclusively through *structure-agnostic* levers. On the optimization side, Grokfast amplifies the slow component of the gradient signal [4], and NeuralGrok learns a gradient transformation [5]. On the parameter side, Truong et al. [6] establish a causal delay law: clamping the total weight norm to a multiple ρ of the critical norm rescales the time-to-grok as $T \propto e^{\alpha\rho}$ ($\alpha \approx 7.5$), with Truong [7] arguing the norm acts through logit scale and softmax saturation. Closest to us, on the representation side, Tiwari et al. [8] penalize the radial component of hidden activations, constraining representations to a hypersphere and accelerating grokking up to $6.3\times$; the intervention is purely geometric, and the authors note they cannot isolate which of several correlated mechanisms drives the effect. None of these interventions carries any information about *which* structure the task requires; whether task-specific structural content is itself a causal lever on the delay is the question we address.

Representation-timing accounts. Chou et al. [9] decompose learning into a fast readout and a slow, continuously improving encoder, and argue — observationally — that grokking occurs when representation quality catches up to an already train-biased readout. Sivasankar [10] report that synchronization of Fourier sub-circuits precedes the accuracy jump, and tas [15] show cross-entropy training converges to task-intrinsic cyclic geometry. Song and Ye [16] frame grokking as a race between memorization and generalization speeds. These accounts collectively predict that the delay measures the time to form task-structured representations, but none intervenes on that timing directly. Our experiment is the interventional test of this prediction.

Supervised contrastive learning. SupCon [11] pulls together representations of same-class examples and pushes apart different-class examples using a temperature-scaled InfoNCE objective over normalized projections. We repurpose it not as an accuracy booster but as a *structure-injection device*: its positive-set definition is a free parameter into which arbitrary equivalence structure can be written — the task’s true structure, or a size-matched random partition — while holding the loss form, strength, and geometry fixed.

3 Method

3.1 Task and model

We use the canonical grokking task: modular addition $c = (a + b) \bmod p$ with $p = 97$, presented as token sequences $[a, b, =]$ over a vocabulary of $p+1$ symbols. Of the $p^2 = 9409$ input pairs, 30% form

the training set and the remainder the test set. The model is a one-layer decoder-only transformer following Nanda et al. [2]: $d_{\text{model}}=128$, 4 attention heads, $d_{\text{mlp}}=512$ with ReLU, learned positional embeddings, no biases, and *no LayerNorm*. Omitting LayerNorm is deliberate: Truong et al. [6] show LayerNorm decouples the weight scale from network function, which would render our weight-norm-matched control (below) uninterpretable. The class logits are read from the final-position residual stream $h \in \mathbb{R}^{128}$ by a linear unembedding. Training is full-batch AdamW ($\eta = 10^{-3}$, $\beta = (0.9, 0.98)$, weight decay 1.0), the standard configuration in which grokking robustly occurs.

3.2 A structure-specific representational prior

We augment the task loss with a supervised-contrastive term on the hidden representation:

$$\mathcal{L} = \mathcal{L}_{\text{CE}} + \lambda \mathcal{L}_{\text{SupCon}}(z, \mathcal{S}), \quad z = \text{normalize}(W_p h), \quad (1)$$

where $W_p \in \mathbb{R}^{64 \times 128}$ is a linear projection head used only by the auxiliary loss, and $\mathcal{L}_{\text{SupCon}}$ is the L^{out} variant of Khosla et al. [11] with temperature $\tau = 0.1$, computed over the full training batch:

$$\mathcal{L}_{\text{SupCon}} = \sum_i \frac{-1}{|P(i)|} \sum_{q \in P(i)} \log \frac{\exp(z_i \cdot z_q / \tau)}{\sum_{a \neq i} \exp(z_i \cdot z_a / \tau)}. \quad (2)$$

The positive-set structure $\mathcal{S}$, which defines $P(i)$, is the experimental variable:

- **True structure:** $P(i)$ contains all training examples sharing example i 's label $c = (a + b) \bmod p$ — the task's real equivalence structure. This uses only the training labels already supplied to the cross-entropy term: no new label information is added, only a *representational format* is demanded.
- **Shuffled structure:** $P(i)$ is defined by a fixed random permutation of the training label vector, drawn once at initialization. This yields pseudo-classes with the exact class-size distribution of the true structure — identical loss form, strength, positive-set cardinalities, and geometry — but with no functional relation to the task.

Any difference between these two conditions is attributable to the structural *content* of the prior, not to the presence of a contrastive term, its optimization pressure, or its geometric side-effects.

3.3 Baselines and controls

Grokfast [4]: the EMA variant ($\alpha = 0.98$, $\lambda_{\text{gf}} = 2$) amplifying the slow gradient component — the standard optimization-side accelerator. **Weight-norm-matched control:** because auxiliary losses perturb the weight-norm trajectory, and the norm causally rescales the grokking timescale [6], we record the per-epoch total weight norm of each true-structure run and replay it onto a pure cross-entropy run with identical initialization, data split, and optimizer state dynamics (per-step global rescaling of all base parameters; optimizer moments preserved), following the matched-counterfactual methodology of Truong et al. [6]. If the auxiliary loss acted through the norm channel, this control would reproduce its effect on the delay.

3.4 Measurements

We record t_{fit} (first epoch with train accuracy ≥ 0.99), t_{gen} (first epoch with test accuracy ≥ 0.95 sustained over three consecutive evaluations), and the delay $t_{\text{gen}} - t_{\text{fit}}$. Representation timing is

Condition	n	Grokked	Median t_{gen}	Median paired ratio	First grok
Baseline	10	10/10	19,950	—	5,500
Grokfast	5	5/5	16,150	0.77	10,300
SupCon-true $\lambda=0.1$	10	8/10	28,725	1.17	5,100
SupCon-true $\lambda=0.3$	10	6/10	33,325	1.22	2,850
SupCon-true $\lambda=1.0$	10	8/10	24,400	0.80	2,000
SupCon-shuffled (all λ)	20	0/20	>50,000	≥ 2.2	—
Norm-matched (all λ)	15	0/15	>50,000	≥ 2.2	—

Table 1: Generalization outcomes within the 50,000-epoch budget. “Median paired ratio” is the per-seed ratio $t_{\text{gen}}^{\text{cond}}/t_{\text{gen}}^{\text{baseline}}$ (values < 1 indicate acceleration; censored runs are conservatively scored at the budget). SupCon-true groks in 22/30 runs versus 0/20 for the exactly-matched shuffled-structure control (Fisher exact $p = 1.3 \times 10^{-7}$) and 0/15 for the weight-norm-matched control ($p = 1.9 \times 10^{-6}$).

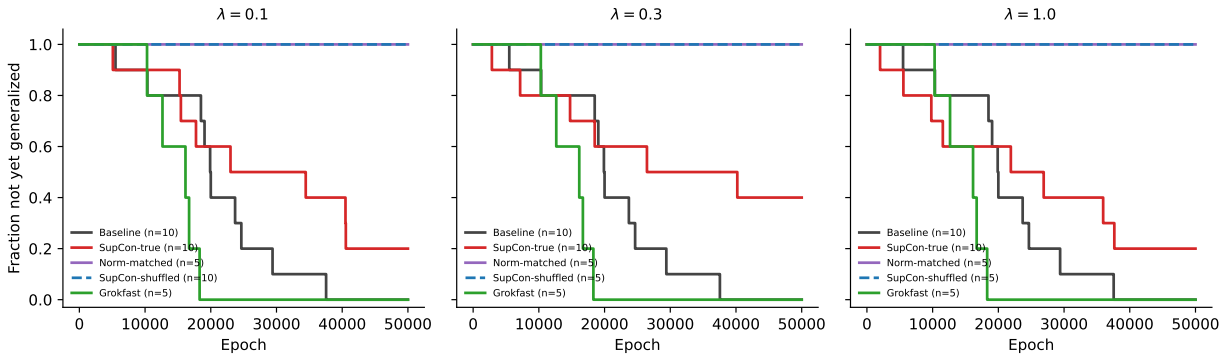


Figure 1: Survival curves: fraction of seeds that have not yet reached 95% test accuracy, per auxiliary-loss strength λ . The true-structure prior (red) produces the earliest generalization in the dataset (2,000 epochs at $\lambda=1.0$, $2.75\times$ faster than the same seed’s baseline) together with a heavy stalled tail; the shuffled-structure control (blue, dashed) and the weight-norm-matched control (purple) never generalize within budget at any λ . Grokfast (green) is a consistent but bounded accelerator.

probed every 50 epochs: (i) the fraction of the number-token embedding’s Fourier power in its top-8 frequencies and its Gini coefficient [2]; (ii) class-clustering of held-out representations — the Fisher ratio (between-/within-class variance) and the cosine gap (mean within-class minus between-class cosine similarity) on a fixed probe set of 1024 test examples; (iii) linear CKA [17] between representations at epoch t and the final post-grokking representations; (iv) the total weight norm and the logit scale, the mediating variables identified by prior work [6, 7]. Statistics use per-seed medians, bootstrap confidence intervals, paired per-seed contrasts, and Fisher exact tests on generalization fractions across ≥ 5 seeds per condition.

4 Results

4.1 Structural content, not the auxiliary loss, controls generalization

Across 30 runs (10 seeds $\times$ 3 strengths), training with the true-structure prior reaches 95% test accuracy within budget in 22 cases (Table 1, Fig. 1). The shuffled-structure control — identical

loss form, strength, temperature, positive-set cardinalities, and normalized-projection geometry — *never* generalizes within budget (0/20; Fisher exact $p = 1.3 \times 10^{-7}$). At $\lambda=1.0$ the contrast is maximal: the true prior yields the fastest generalization we observe anywhere in the dataset (2,000 epochs, versus 5,500 for the same seed’s baseline), while the shuffled prior leaves test accuracy at chance ($\leq 2\%$) and even delays training-set interpolation to $\sim 10\text{--}15\text{k}$ epochs. Whatever the contrastive term contributes — optimization pressure, representation-norm control, gradient noise — is present in both conditions; only the *content* of the equivalence structure differs.

4.2 The effect is not mediated by the weight norm

The true-structure prior inflates the total weight norm from ~ 50 to 105–150, and Truong et al. [6] show the norm alone rescales the grokking timescale exponentially. Replaying each SupCon-true run’s per-epoch norm trajectory onto plain cross-entropy training (same seed, initialization, and split) reproduces the *pathology* but none of the *benefit*: 0/15 norm-matched runs generalize. Pure CE at the inflated norm collapses into saturated memorization — logit scale grows to $\sim 10^4$, softmax confidence exceeds 0.99, test cross-entropy diverges ($> 2,000$), and embedding Fourier concentration never leaves its memorization value (Fig. 2) — exactly the norm $\rightarrow$ logit-scale $\rightarrow$ saturation chain of Truong [7]. Critically, this holds even for the trajectory of the run that grokked in 9,800 epochs (twice the baseline speed): the norm signature of acceleration confers no acceleration. Both contrastive conditions, true and shuffled, avoid the saturation collapse (confidence $\approx 0.6\text{--}0.7$ throughout), identifying a second, structure-*agnostic* channel of the auxiliary loss: it keeps logits soft where decoupled weight decay alone cannot. But anti-saturation without correct structure never yields generalization (shuffled: 0/20).

4.3 A bidirectional, dose-dependent lever with a stochastic trap

The true prior does not uniformly accelerate: per-seed paired ratios at $\lambda=1.0$ span 0.36 (a $2.75\times$ speedup) to complete trapping beyond budget, with median 0.80 (8/10 grokking; 6/10 faster than their paired baseline, i.e. 6 of the 8 that grokked). Acceleration is dose-monotone on seeds that grok (e.g. seed 6: $5,100 \rightarrow 2,850 \rightarrow 2,000$ epochs for $\lambda = 0.1 \rightarrow 0.3 \rightarrow 1.0$), while the stall probability is roughly dose-independent (2/10, 4/10, 2/10). Stalled runs are not dead: their test accuracy and Fourier concentration climb slowly (e.g. 28% and 0.54 at budget), indicating a delayed, not destroyed, transition. Trajectory analysis suggests a race: seeds that develop class-clustered representations before norm inflation saturates the readout ride the injected structure to early generalization; seeds that miss this window inherit the high-norm slow regime. Under matched seeds, Grokfast [4] is a consistent but bounded accelerator (5/5 grokked, ratios 0.43–0.85); the structural prior reaches larger peak speedups at the cost of the trap mode.

4.4 Representation timing explains generalization timing

The probes tie the delay to the timing of structure formation. In the true-structure condition the class cosine gap of held-out representations rises during the plateau, thousands of epochs before the accuracy jump, and embedding Fourier concentration rises earlier than in any other condition (Fig. 2); the accuracy transition occurs as these measures complete their ascent. In the shuffled condition the same optimization pressure produces *no* task-relevant class structure (cosine gap ≈ 0 until late), and in the norm-matched condition no structure forms at all. Across all 80 runs, none generalizes before its representation probes move, and none completes the Fourier rise without

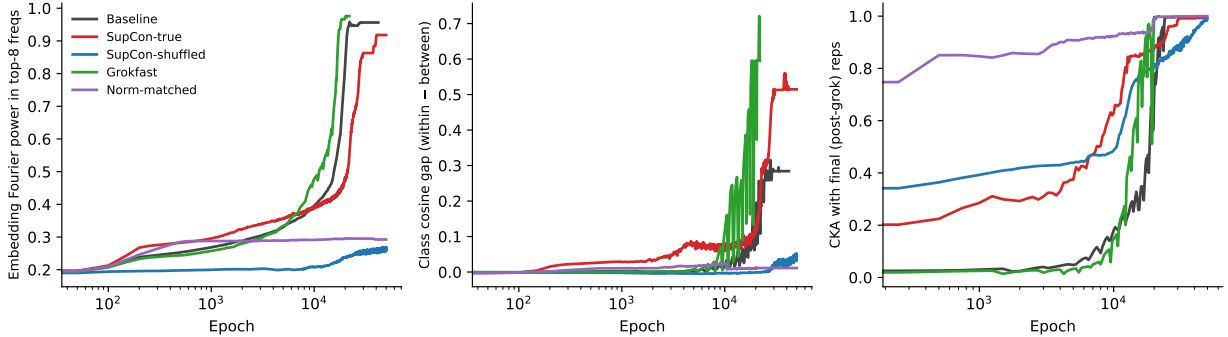


Figure 2: Representation-timing probes (medians across seeds, primary $\lambda=1.0$). **Left:** fraction of embedding Fourier power in the top-8 frequencies. **Middle:** class cosine gap (within-class minus between-class mean cosine similarity) of held-out representations. **Right:** linear CKA between representations at epoch t and the same run’s final representations. The true-structure prior (red) builds class structure during the apparent plateau, ahead of its own generalization; the norm-matched control (purple) never forms structure (its high CKA reflects representations frozen at memorization); the shuffled control (blue) shows no task-structure formation despite equal optimization pressure.

generalizing¹ — the interventional counterpart of the observational representation-first accounts [9, 10].

5 Discussion

The delay is representation-limited, causally. The representation-first accounts of grokking — slow encoder, fast readout [9], circuit synchronization preceding the jump [10], structured-embedding quality gating generalization [12] — predicted exactly the intervention outcome we observe: demand the task’s equivalence structure in representation space and generalization time shrinks, down to $2.75\times$ faster than baseline; demand an equally strong but wrong structure and generalization never arrives. The grokking delay in this system is therefore not an unavoidable optimization constant but the time for task-structured representations to form, and it can be manipulated in either direction by controlling which structure forms.

Reconciling with the weight-norm delay law. Our results do not contradict Truong et al. [6]; they bound its scope. The norm channel is real and strong in our data: inflating the norm of a plain-CE learner to the SupCon trajectory reliably destroys generalization within budget (0/15), via the logit-scale saturation mechanism of Truong [7]. But the delay law is a law for *structure-agnostic* training. A structure-specific objective breaks it in both directions: SupCon-true generalizes at norms (105–150) where plain CE cannot, and its acceleration survives on trajectories whose norm signature alone confers none. Timescale control by the norm is thus conditional on the absence of stronger representational forces — a boundary condition future delay-law work must state.

¹Operationalized as: every run reaching 95% test accuracy first crossed cosine gap > 0.05 or Fourier top-8 fraction > 0.45 at an earlier epoch (80/80 satisfied); no run reached Fourier top-8 > 0.8 without generalizing (80/80).

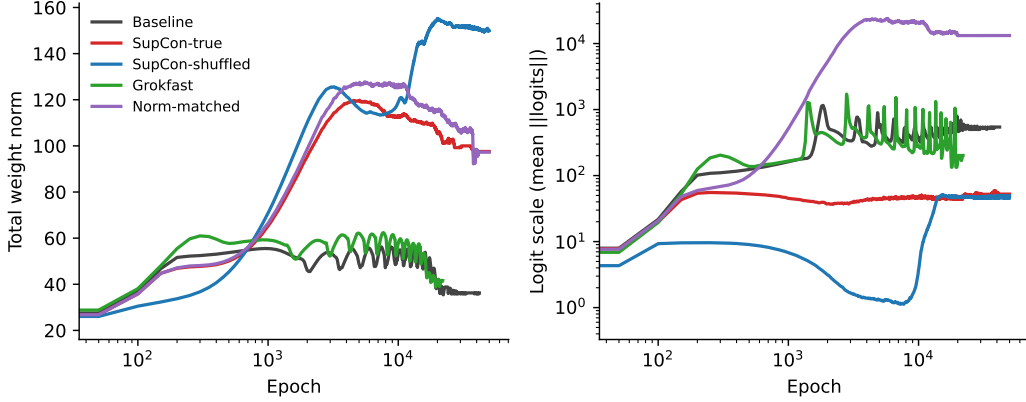


Figure 3: Mediating variables (medians across seeds, primary $\lambda=1.0$). **Left:** total weight norm. Both contrastive conditions inflate the norm well above the baseline’s weight-decay equilibrium; the norm-matched control (purple) replays the SupCon-true trajectory by construction. **Right:** logit scale. Pure cross-entropy at the inflated norm diverges into softmax saturation (purple), while both contrastive conditions hold logits soft — the structure-agnostic anti-saturation channel. Neither variable separates true from shuffled structure; only the representational content does.

Geometry versus content. Tiwari et al. [8] accelerate grokking by penalizing radial inflation of hidden activations and note they cannot isolate which of several correlated geometric mechanisms is operative. Our shuffled control speaks to this: a loss with identical normalized-projection geometry, identical strength, and identical anti-saturation behavior, differing only in the partition it encodes, produces zero generalization. Geometric regularization of representations can be a useful lever, but in our setting geometry without content never suffices — and content overrides adverse geometry (norm inflation). The two families of intervention are mechanistically distinct, and only the structural one is bidirectional.

The trap mode and the race. The failure mode of the true-structure prior is as informative as its success. Injected structure competes with a side-effect of its own optimization: contrastive gradients on normalized projections inflate upstream weight norms, which drives the readout toward saturation. Seeds whose class clusters consolidate early convert the prior into early Fourier structure and grok up to $2.75\times$ sooner; seeds that miss the window inherit a slowed, high-norm regime — though still one qualitatively better than its norm-matched counterfactual (climbing versus flat test accuracy). This race picture predicts that stabilizing the norm side-effect (e.g. norm-preserving contrastive gradients, or annealing λ) would retain the acceleration while removing the trap; we leave this to future work.

Practical note. Grokfast remains the more *reliable* accelerator in this setup (5/5, $1.2\text{--}2.3\times$); structure injection achieves larger peak speedups ($2.75\times$) with stall risk. The two act on different variables (gradient spectrum versus representation target) and are in principle composable.

6 Limitations

Our evidence comes from one task family (modular addition, $p = 97$) and one small architecture, the regime where grokking is cleanest and where the mechanistic ground truth (the Fourier circuit)

is known; transfer to larger models and natural data is untested, a limitation shared with the interventional grokking literature [6, 8]. The structure-specific prior consumes only information already present in the training labels, but it is supervised: a self-supervised variant (positives from algebraic invariances rather than labels) would sharpen the claim. The shuffled control matches class sizes, loss form, and strength, but a fixed random partition is also *learnable* by memorization; it controls for optimization pressure, not for the learnability of the partition. Finally, the contrastive term adds $O(n^2)$ similarity computation per step, roughly $1.5\times$ wall-clock in our full-batch setting; we compare conditions in epochs, not wall-clock.

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A Hyperparameters and reproducibility

Task	$(a + b) \bmod 97$, tokens $[a, b, =]$, 30% of 97^2 pairs train
Model	1-layer transformer, $d_{\text{model}}=128$, 4 heads, $d_{\text{mlp}}=512$ (ReLU) no LayerNorm, no biases, learned positional embeddings
Optimizer	full-batch AdamW, $\eta=10^{-3}$, $\beta=(0.9, 0.98)$, weight decay 1.0
SupCon	L^{out} variant, $\tau=0.1$, linear projection $128 \rightarrow 64$ (no bias), $\lambda \in \{0.1, 0.3, 1.0\}$, computed on the full training batch
Grokfast	EMA filter, $\alpha=0.98$, $\lambda_{\text{gf}}=2.0$
Norm-matched	per-step global rescale of base parameters to the recorded per-epoch norm of the paired (same-seed) SupCon-true run
Budget	50,000 epochs cap; early stop 3,000 epochs after sustained test acc ≥ 0.99
Thresholds	t_{fit} : train acc ≥ 0.99 ; t_{gen} : test acc ≥ 0.95 for 3 consecutive evals
Seeds	≥ 5 per condition, shared across conditions (matched init and split)
Hardware	single NVIDIA RTX 3080; full grid ≈ 4 GPU-hours

Table 2: Complete configuration. Code and run artifacts accompany the paper.

B Additional figures

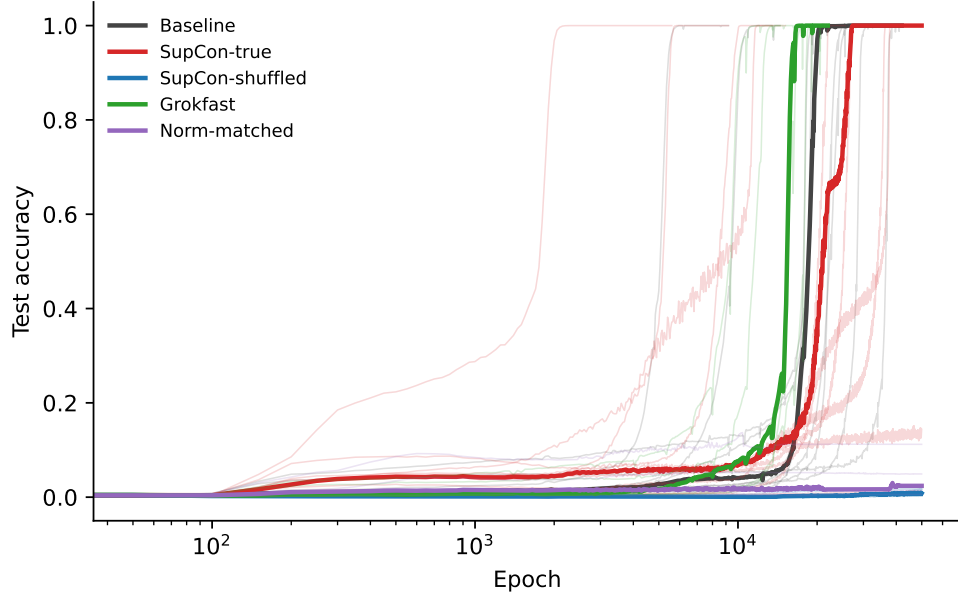


Figure 4: Test-accuracy trajectories at the primary strength $\lambda=1.0$ (thin: individual seeds; thick: medians). Median curves for conditions with stalled runs reflect the bimodal outcome distribution.

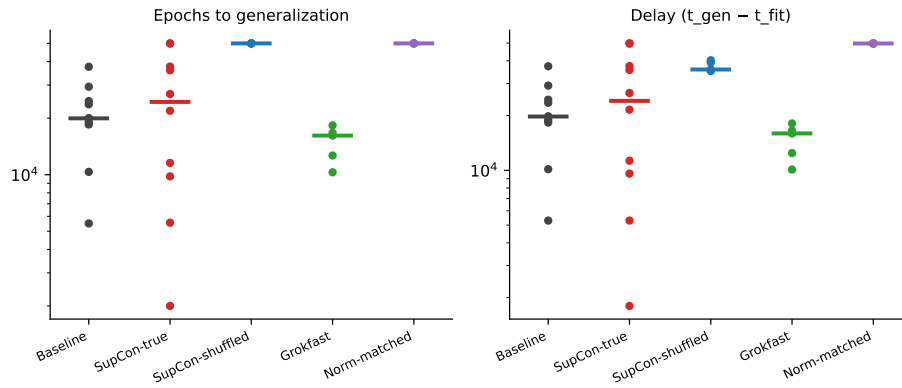


Figure 5: Epochs to generalization and delay ($t_{\text{gen}} - t_{\text{fit}}$) per condition at $\lambda=1.0$ (points: seeds; bars: medians; censored runs plotted at the 50,000-epoch budget).

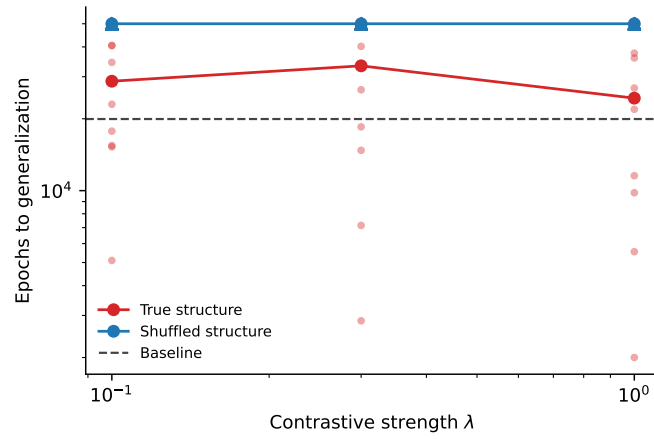


Figure 6: Dose–response: epochs to generalization versus auxiliary-loss strength λ (triangles: censored runs at budget; line: medians). True structure (red) spans strong acceleration to trapping; shuffled structure (blue) never generalizes at any strength.